

# OPEN RAIL

## Jing-Zhang AI Commons • Urban Design Proposal (A3 Booklet)

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Open call: Centennial Jing-Zhang AI Innovation Belt open call

Iteration: v2.8 • 2026-08

Nature: Concept proposal / reference scheme; not a statutory plan or government conclusion

All areas and ratios in this booklet are recomputable from package GeoJSON in EPSG:4548. Boundaries are provisional; full recalculation is required upon official data release.

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## 01 Design Basis & Sources

The primary authority is the official pre-qualification announcement [source:OFFICIAL-ANNOUNCEMENT]; followed by the public site package [source:SITE-PACKAGE] and the agent taskbook [source:AGENT-TASKBOOK]; cultural materials serve narrative only [source:JINGZHANG-HERITAGE-PARK]. Source usability follows the public registry [source:SOURCE-REGISTRY].

| Source                    | Level                 | Usage                           |
|---------------------------|-----------------------|---------------------------------|
| Official announcement     | official_public       | Tasks, scopes, areas            |
| Site package              | official_public       | Structured brief & rules        |
| Agent taskbook            | user_provided_cleared | Six tasks & boundary clause     |
| Provisional boundaries    | provisional_only      | Interim generation; not redline |
| Heritage park public info | background_only       | Narrative background            |

## 02 Three-Level Scope & Boundary

| Level         | Area                 | Extent                                                          |
|---------------|----------------------|-----------------------------------------------------------------|
| Research area | 43.6 km <sup>2</sup> | N 5th Ring / Jingzang Expwy / Xizhimenwai / Wanquanhe           |
| Design area   | 11.4 km <sup>2</sup> | 1–2 km around heritage park (recomputed 11,412,825 sqm, +0.11%) |
| Key areas     | 368.4 ha             | Zhongzhiyuan 192.1 / AI Origin 104.3 / Dazhongsi 72.0 ha        |

Official redlines are unpublished; all geometry derives from provisional\_boundaries.geojson (provisional\_rough) and is not for approval or precise area use.

### 03 Existing Conditions & Gap List

- The heritage corridor runs north–south but splits east–west links; six crossings have poor walking experience.
- Campus, community and park frontages are closed; innovation flows rely on arterial roads.
- Qing River and Xiaoyue River waterfronts lack systematic connection with the corridor.
- AI scenarios lack accessible, testable, displayable public-space carriers.

#### Data gaps (pending official data)

- Official precise boundary & key-area polygons
- Current regulatory controls (FAR, height, density)
- Existing buildings, property rights, municipal capacity

### 04 Research Area: Industry & Future-City Strategy

**Concept:** OPEN RAIL • Jing-Zhang AI Commons: the 9.7 km corridor as civic mainboard; rails as data & slow-mobility bus; stations as compute-community nodes; three areas as chips.

Three positionings form a timeline trio; five functions are anchored: full-stack to Zhongzhiyuan, ecosystem to Origin Community, scenarios to spine + Xiaoyuehe wing, vitality city to the slow network, governance voice to the testbed.

Six global cases distill transferable mechanisms of corridor-as-ecosystem, heritage-as-scenario, station-as-service [source:GLOBAL-AI-ECOSYSTEM-CASES].

#### Regional synergy (proposals, pending negotiation)

| Partner              | Exchangable resources                                   | Interface & verification                                      |
|----------------------|---------------------------------------------------------|---------------------------------------------------------------|
| Beiwei Community     | Developer community, open-source branding, young talent | Wudaokou interface + honour-wall recognition; quarterly stats |
| Future Science City  | Compute & energy experience, SOE R&D talent             | Zhongzhiyuan compute access; joint-lab agreements             |
| Huairou Science City | AI-for-Science demand, compliant data cooperation       | Origin annual exhibition; shared filing tools                 |
| Beijing E-Town       | Embodied-AI lines, robot supply chain                   | Dazhongsi front store; joint test reports                     |
| Jing-Jin-Ji region   | Regional compute network, talent commuting & housing    | Qinghe compute access; annual housing review                  |

All synergies are conceptual proposals requiring partner agreement; no government commitments presumed.

## 05 Overall Design: Structure & Land Use

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Land-Use Structure: spine as axis, parcels as clusters

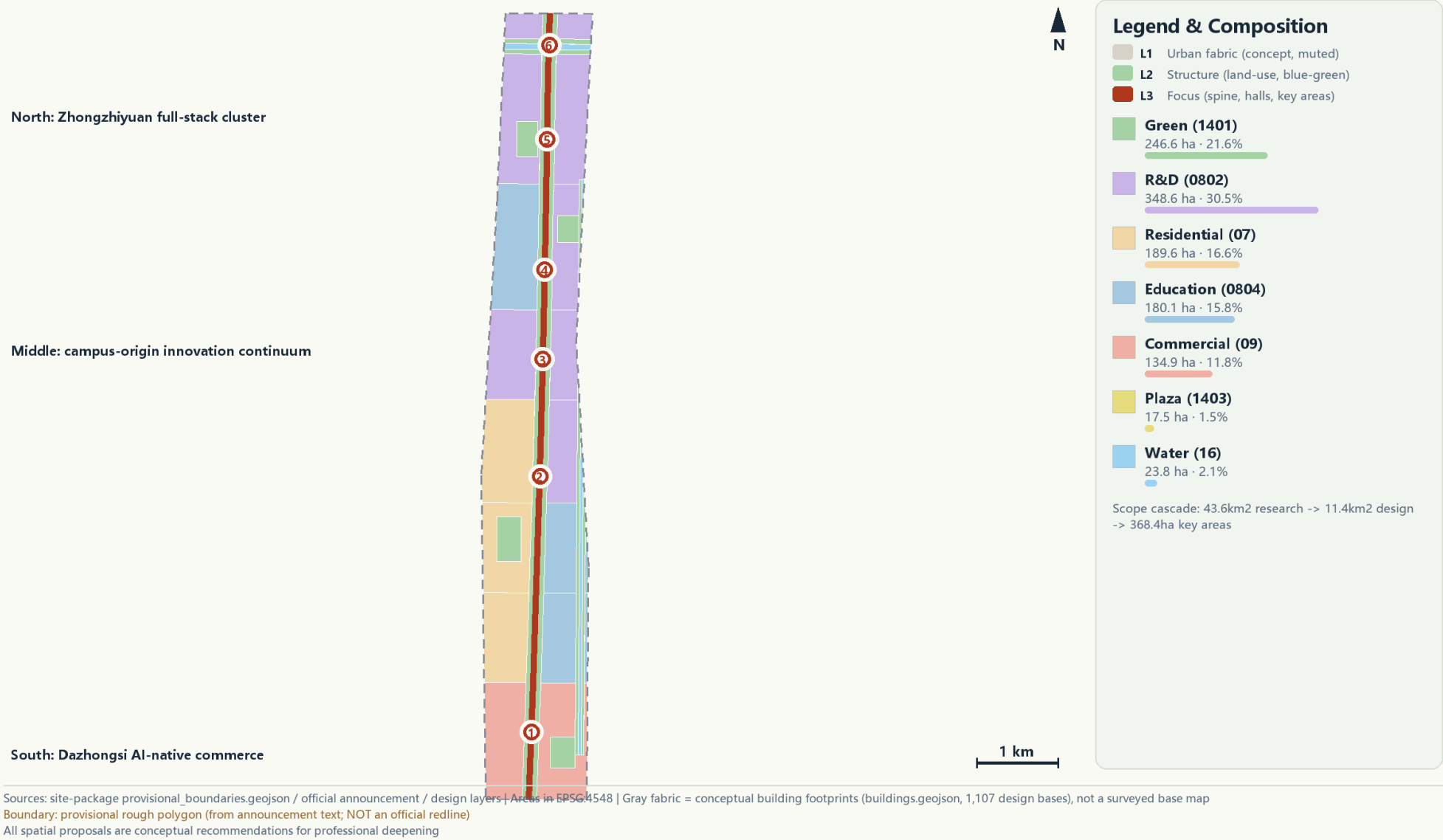


Fig.1 | Land-use structure (conceptual, provisional boundary)

| Use            | Area     | Share |
|----------------|----------|-------|
| R&D 0802       | 348.6 ha | 30.5% |
| Green 1401     | 246.6 ha | 21.6% |
| Residential 07 | 189.6 ha | 16.6% |
| Education 0804 | 180.1 ha | 15.8% |
| Commercial 09  | 134.9 ha | 11.8% |
| Water/plaza    | 41.4 ha  | 3.6%  |

Structure: 180 m corridor as spine, six halls as joints, three areas as chips, two wings as interfaces.

# 05b Overall Design: Mobility & Blue-Green

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Mobility x Blue-Green: stitched corridors, riverfront pearls

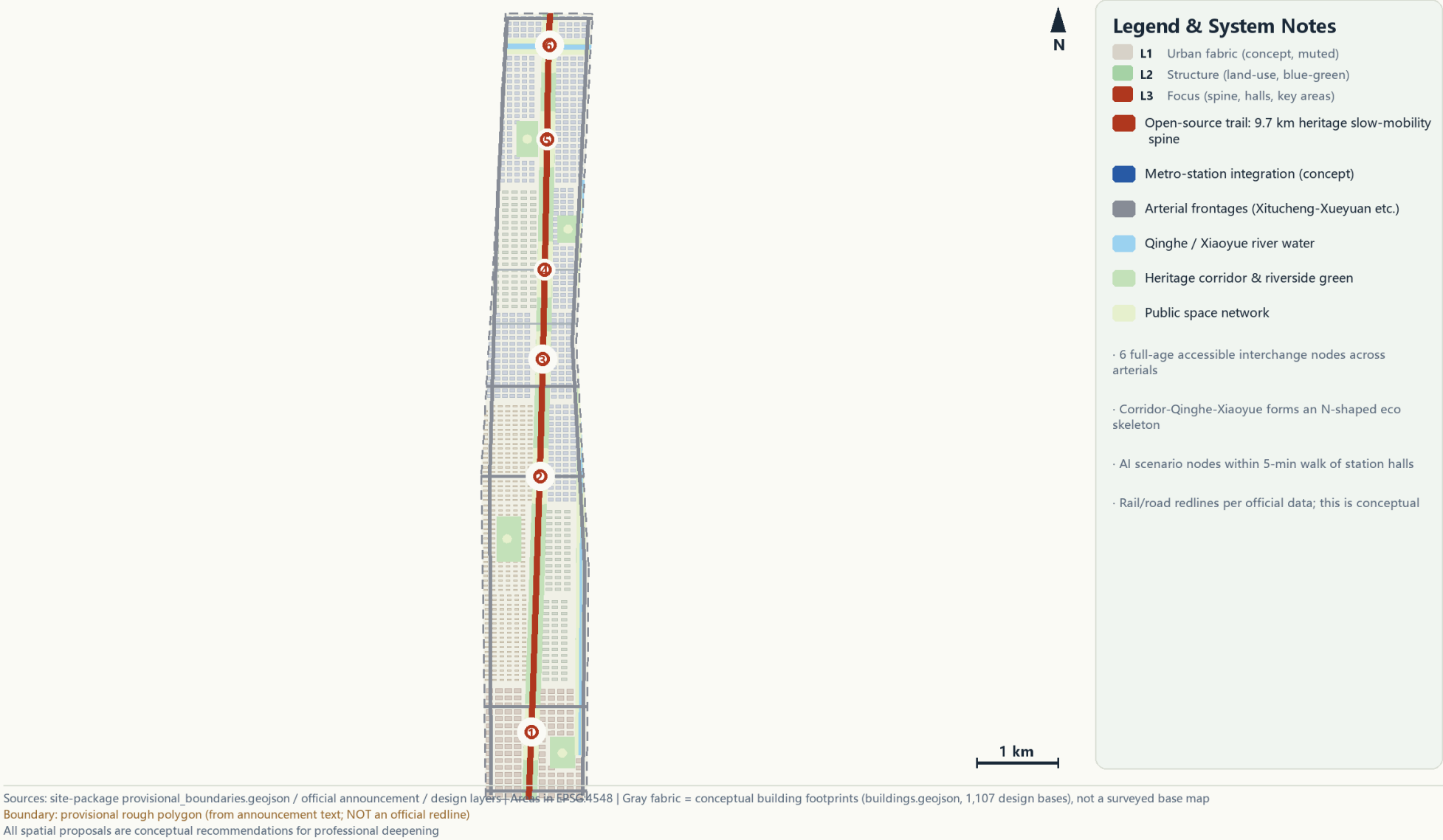


Fig.2 | Mobility and blue-green system (conceptual)

- 9.72 km trail runs north–south; six halls stitch east–west; metro integration is conceptual.
- Green ratio 21.61% and public-space ratio 10.56% (design-model, recomputable).
- Edge-compute nodes at six halls; distributed energy and sponge facilities along riverside green.

06 Key-Area Detailed Design

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Three Key Areas: positioning, anchors, risk conditions

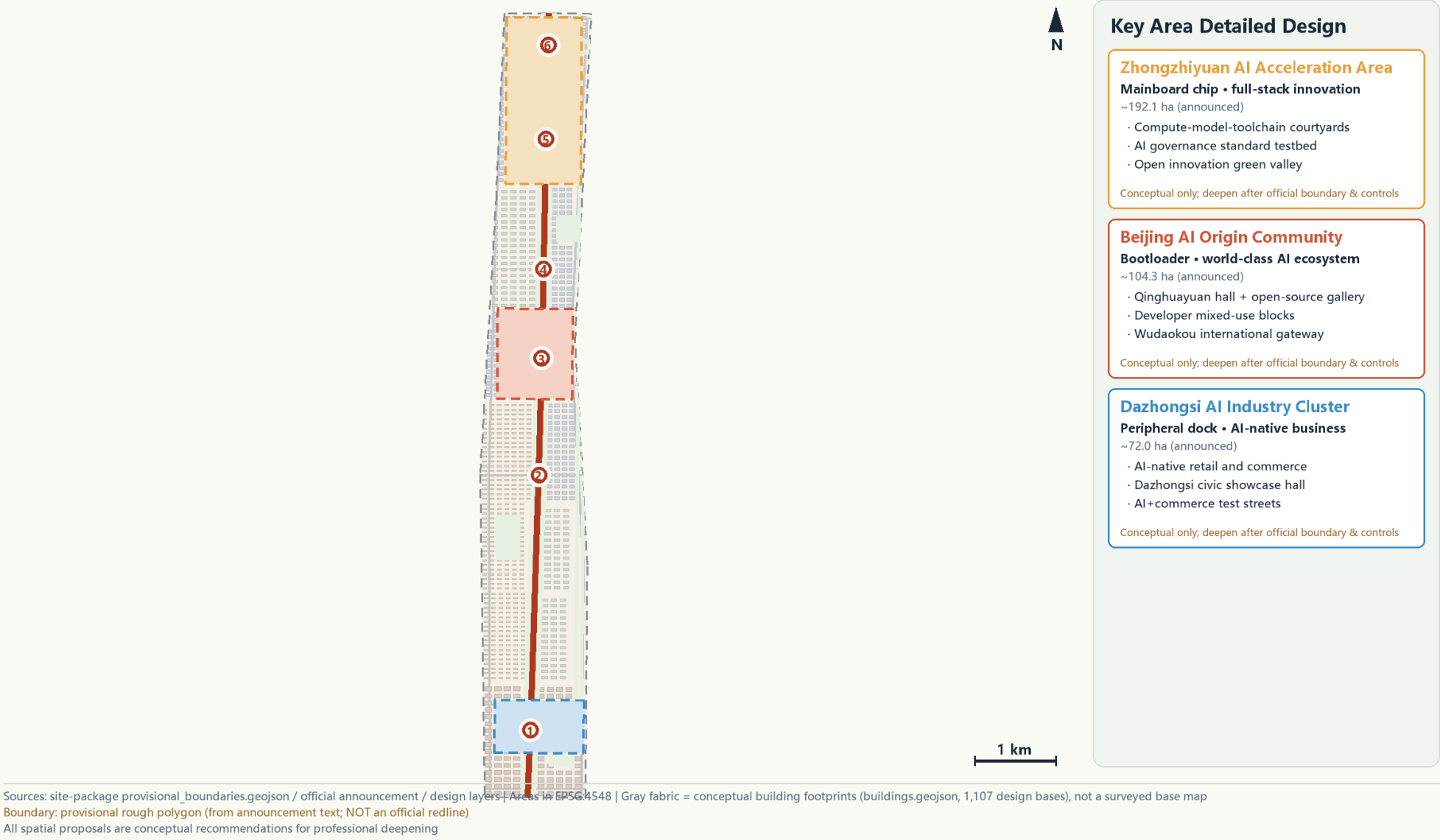


Fig.3 | Key areas and design tasks (conceptual)

| Area                 | Role                                 | Anchors                                                              |
|----------------------|--------------------------------------|----------------------------------------------------------------------|
| Zhongzhiyuan 192.1ha | Mainboard • full-stack               | Compute-model-toolchain courtyards, governance testbed, green valley |
| AI Origin 104.3ha    | Bootloader • ecosystem               | Qinghuayuan gallery, developer blocks, Wudaokou gateway              |
| Dazhongsi 72.0ha     | Peripheral dock • AI-native commerce | AI-native retail, civic showcase, AI+commerce tests                  |



06b Place Studies (1): Halls & Brand

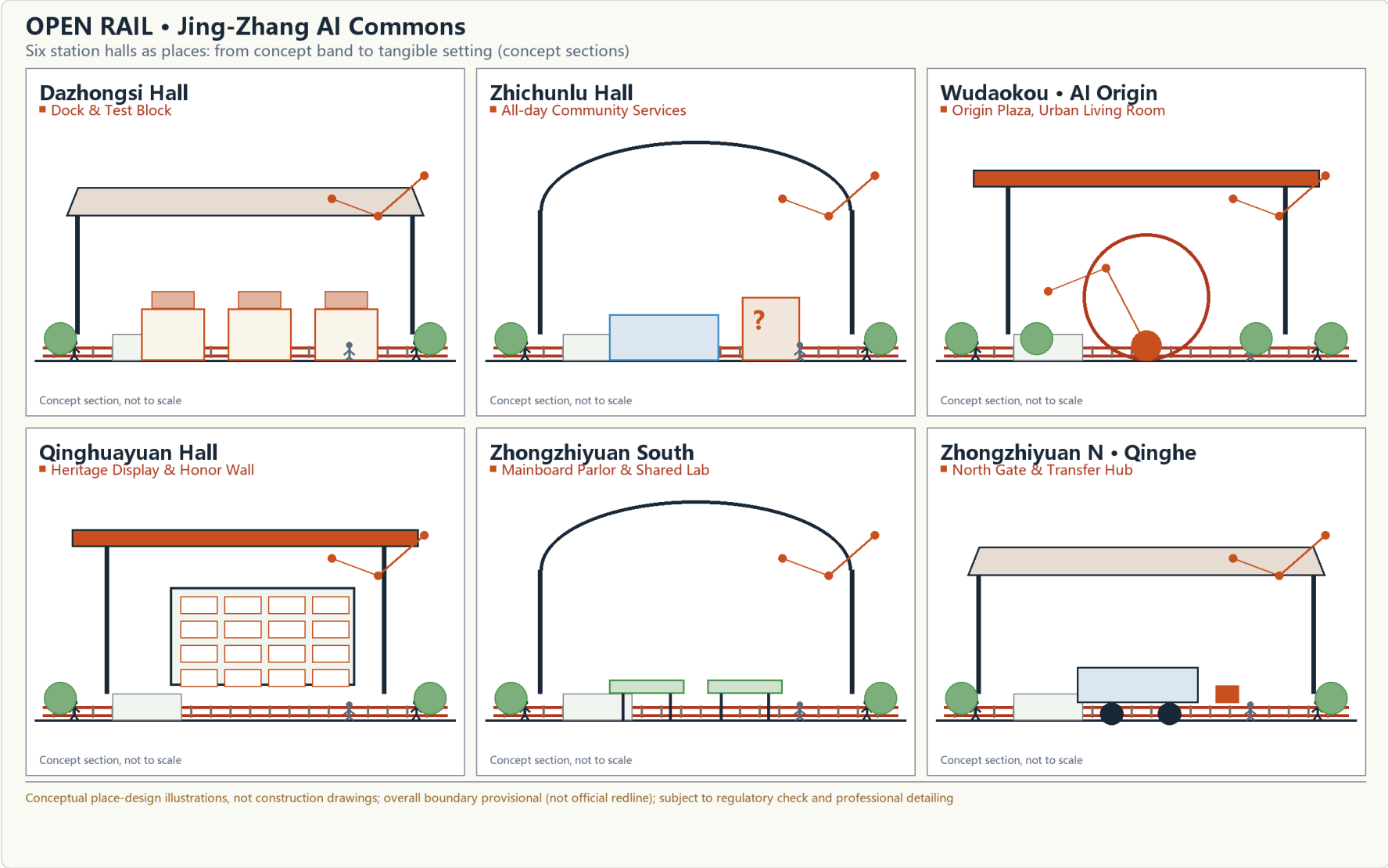


Fig.3b | Six halls as places (concept sections, not construction drawings)

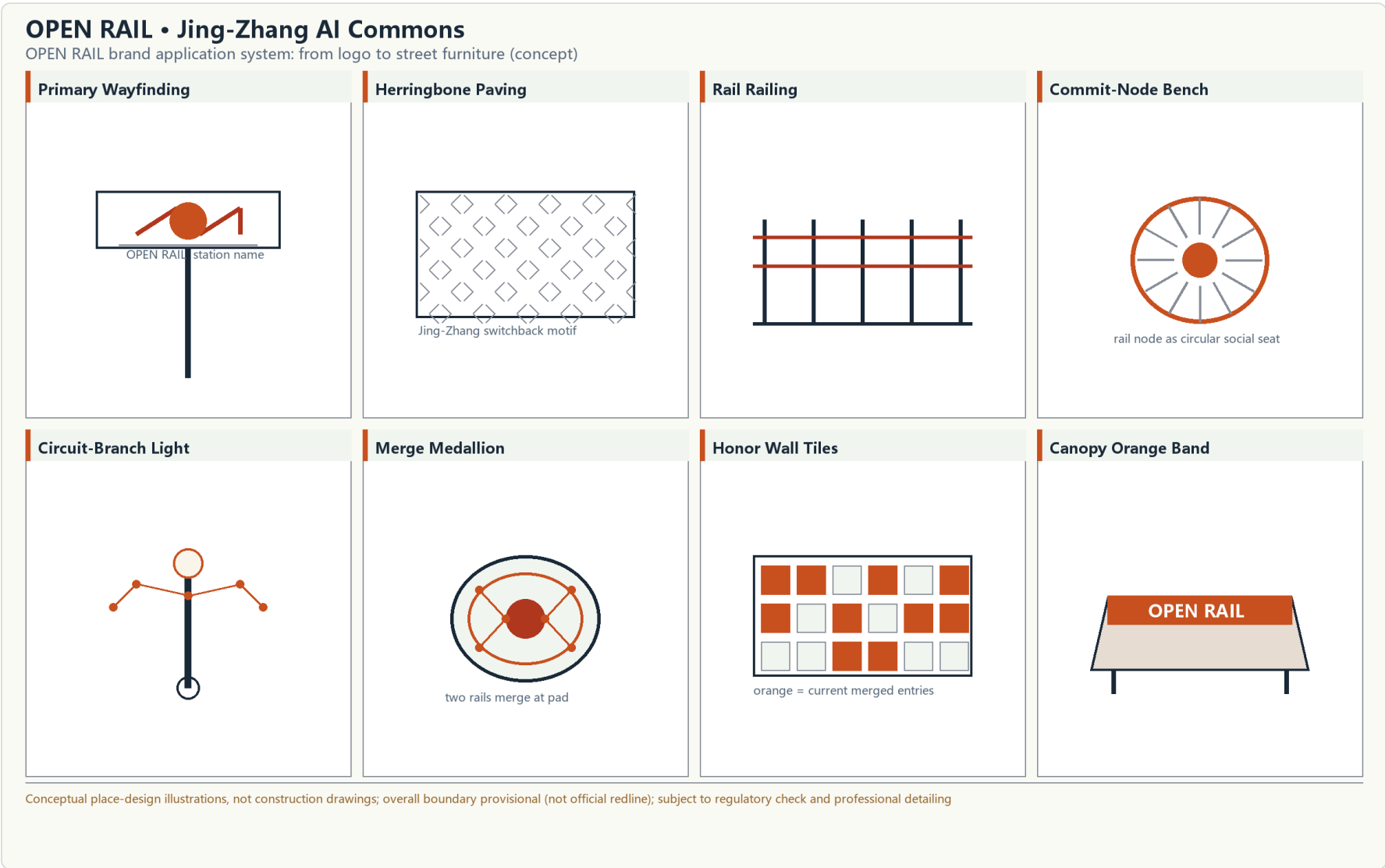


Fig.3c | OPEN RAIL brand application system

# 06b Place Studies (2): Commit Plaza Node

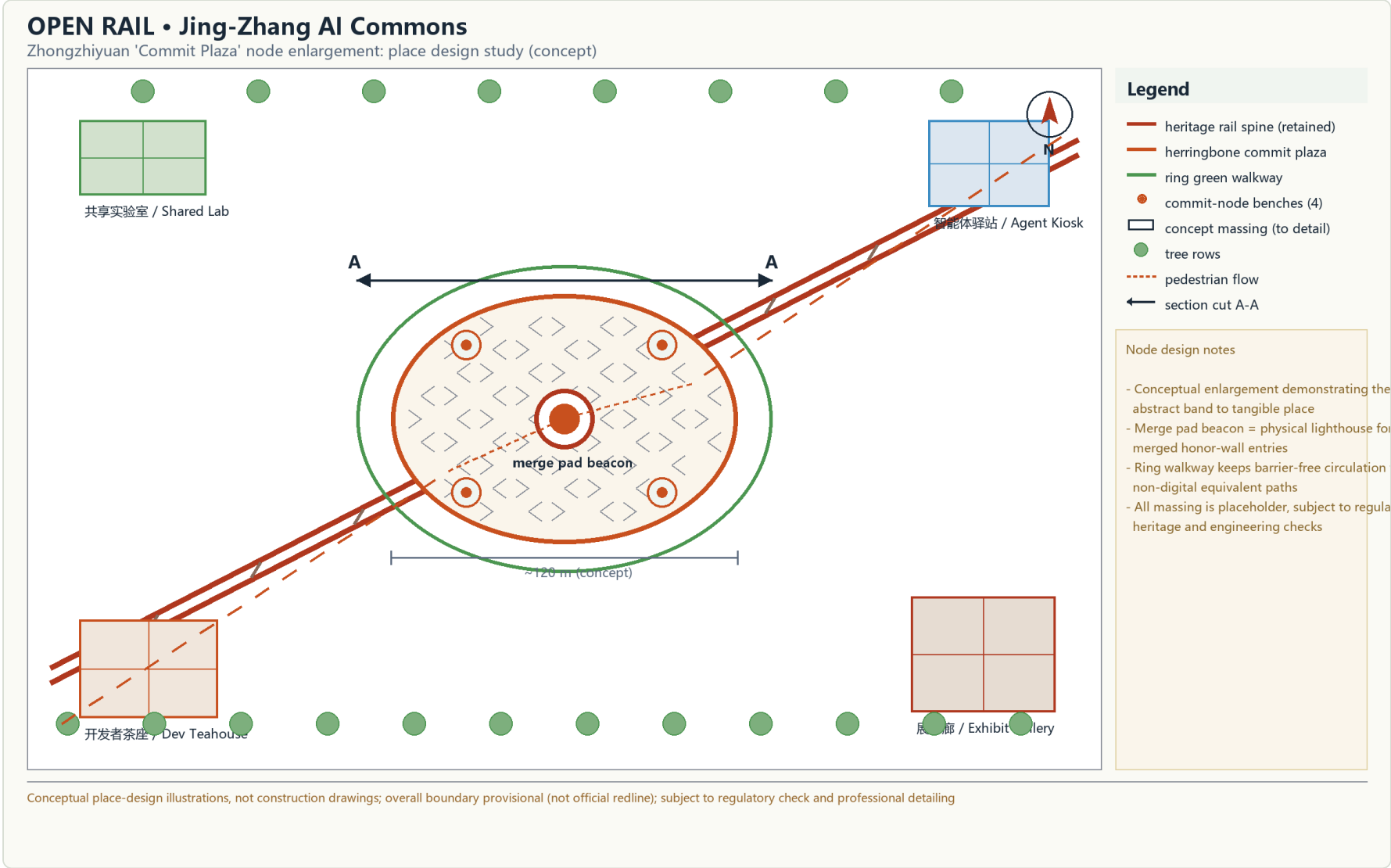


Fig.3d | Commit Plaza node enlargement (placeholder massing)

From abstract band to tangible place: hall sections, branded furniture and a node enlargement demonstrate the direction. All are concept studies subject to regulatory, heritage and professional checks.



# 06c Streetfront & Construction Studies

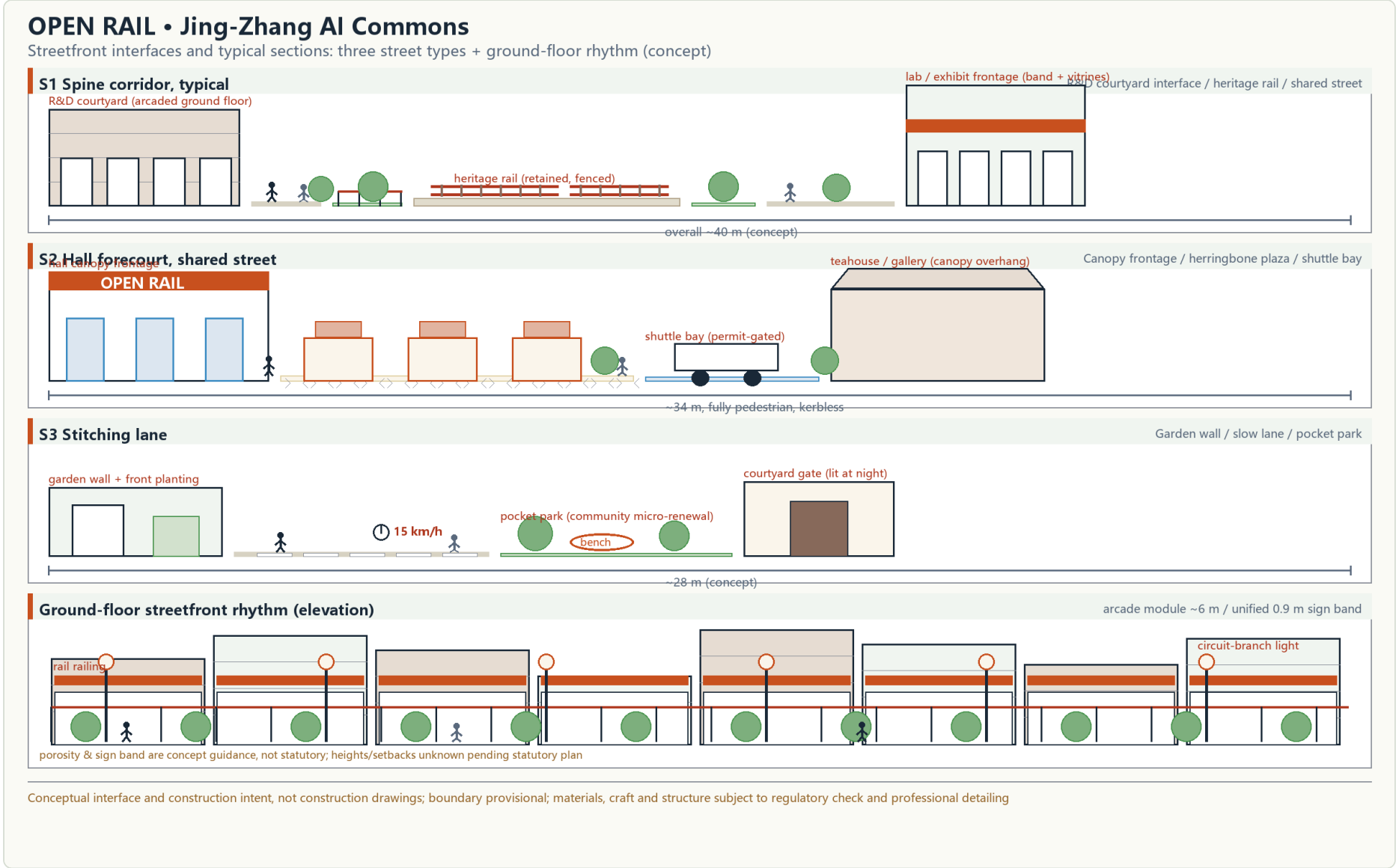


Fig.3e | Streetfront interfaces & typical sections (S1 corridor / S2 forecourt / S3 lane + ground-floor rhythm; concept)

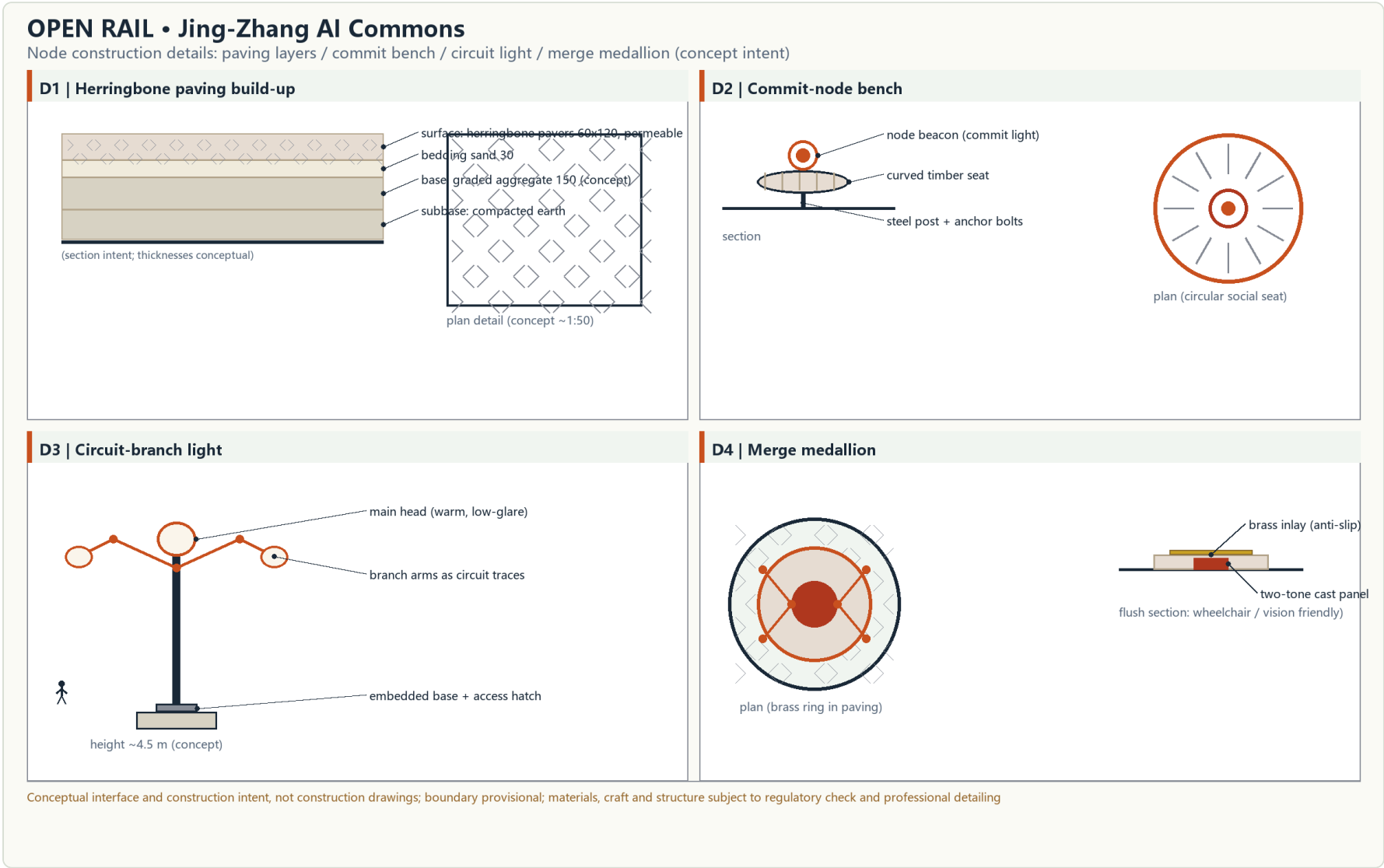


Fig.3f | Node construction details: paving / bench / light / medallion (intent, not construction drawings)

Heights/setbacks unknown pending statutory plan; materials and structure follow professional detailing and accessibility review.

# 07 AI+ Scenarios & Data Governance

13 scenario cards all carry data boundaries and human review; 3 test scenarios require authority permission:

AI+walkability

AIcultural guide

AIhealth nav

enterprise copilot

safety review

test: robot delivery

AI+open class

AI+retail guide

edge-compute test

smart park O&M

test: autonomous shuttle

test: governance sandbox

AI accessibility companion (non-digital equivalents)

## Six personas

Researchers / developers / students & faculty / residents / visitors / older adults & accessibility needs — each mapped to space supply, scenarios and operations.

## AI pilgrimage landmarks (4 concepts)

Developer Promenade

Agent Honour Wall

Open-Source Gallery

AI Origin Monument

## Privacy & governance

Personal data stays on device, authorised use, human review; no excessive surveillance; tests are segment-limited, speed-limited, takeover-ready.  
Scenario governance: component-level adoption of community open standards — Switchback Protocol v1.0 and SEB v0.5.0 node schema; non-official methods, not statutory bases.

## 08 Renewal Projects & Phasing

| Phase        | Scope                         | Key projects |
|--------------|-------------------------------|--------------|
| Near 2026–29 | S. spine + Origin + Dazhongsi | 1,2,3,5      |
| Mid 2029–32  | N. corridor + Zhongzhiyuan S  | 1,4,6,7      |
| Far 2032–35  | Zhongzhiyuan N + Qinghe       | 7,8          |

Projects: 1 spine link; 2 six halls; 3 Qinghuayuan gallery; 4 green valley; 5 Dazhongsi renewal; 6 housing + pocket parks; 7 riverside link; 8 edge-compute demo. Bodies, funding and approvals follow formal decisions.

### Annual events & operations (concept)

Spring conference / summer test season / autumn culture season / winter agent annual; GitHub PR/Issue as standing co-creation routine.

### Funding typology & entity paths (proposals)

| Type                  | Applies to             | Entity path                  | Boundary                   |
|-----------------------|------------------------|------------------------------|----------------------------|
| City/district funds   | 1&7 public works       | District projects, statutory | Per formal approval        |
| Market investment     | 5 blocks, courtyards   | Concession/co-development    | Property negotiation first |
| Industry funds        | Scenarios & incubation | Open scenarios + matchmaking | No promises presumed       |
| Foundations           | Cultural content       | Charter, disclosed           | Heritage review first      |
| Community micro-funds | 6 stitching nodes      | Participatory budgeting      | Consent threshold          |
| Operations reinvest   | Events & operations    | Disclosed under review       | Audit scope TBD            |

### Baseline protocols (pre-launch)

P1 accessibility coverage (quarterly) / P2 equivalent paths / P3 response median / P4 walkability breaks / P5 third-party audit; failures retained; unmeasured stays pending-baseline.

### Sequencing & cost-benefit methodology (proposal, trigger-gated)

Step 1 data baseline (on official geometry/statutory release) / 2 priority scoring (weighted, negotiable) / 3 cost lines (no estimates) / 4 benefit lines (P1-P5 + proxies) / 5 sensitivity (robust intervals). No ranking or benefit figures before all triggers; conclusions jointly reviewed and rolled with the 90-day review.

09 Metrics, Recalculation & Compliance

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Metrics & Evidence: recomputable from package geometry

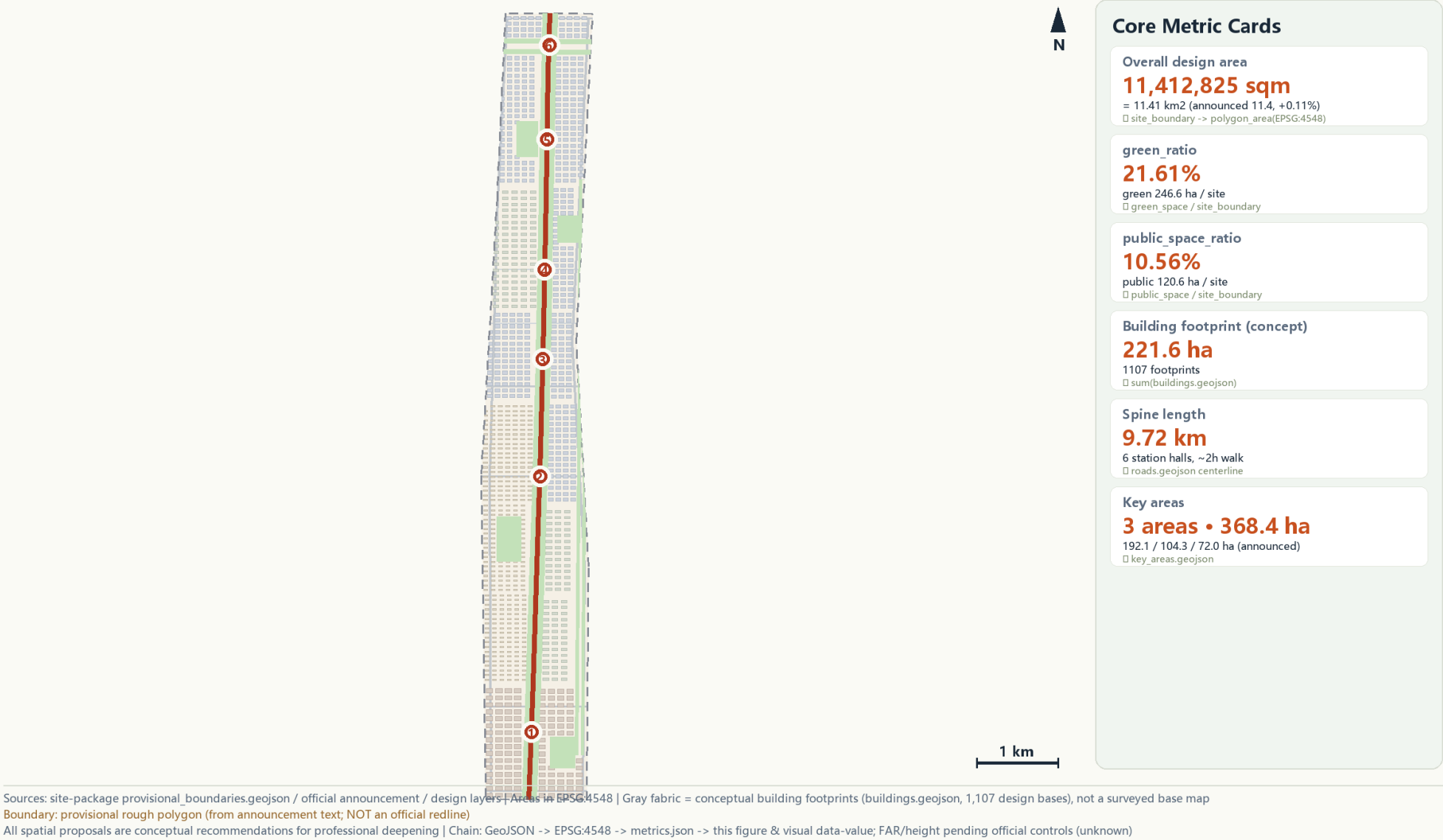


Fig.4 | Metrics recalculation and evidence chain

|                                                              |                                |                               |
|--------------------------------------------------------------|--------------------------------|-------------------------------|
| <b>11,412,825</b><br>m <sup>2</sup> design area (recomputed) | <b>21.61%</b><br>green ratio   | <b>10.56%</b><br>public space |
| <b>221.6 ha</b><br>footprints (concept)                      | <b>9.72 km</b><br>spine length | <b>368.4 ha</b><br>key areas  |

FAR/height: unknown pending controls. Compliance matrix covers all 23 requirements (1.3/1.4/1.5 + agent.1-6); standard matrix covers mandatory standards; all 15 depth items complete.

## 10 Risk, Copyright & Missing Data

- Provisional boundary: full recalculation upon official data (A-BOUNDARY-002).
- Regulatory gap: intensity/height/demolition pending (A-CONTROLS-001).
- Heritage: Qinghuayuan content is conceptual, subject to relics protection.
- Implementation: phasing and projects are advisory, not commitments.

Generated by an AI agent from public/cleared sources; figures are original programmatic renderings. See report/copyright\_statement.md.

### Bilingual

This booklet has an English counterpart; proposal.md (zh) and proposal.en.md (en) mirror chapters, claims, metrics and figures.